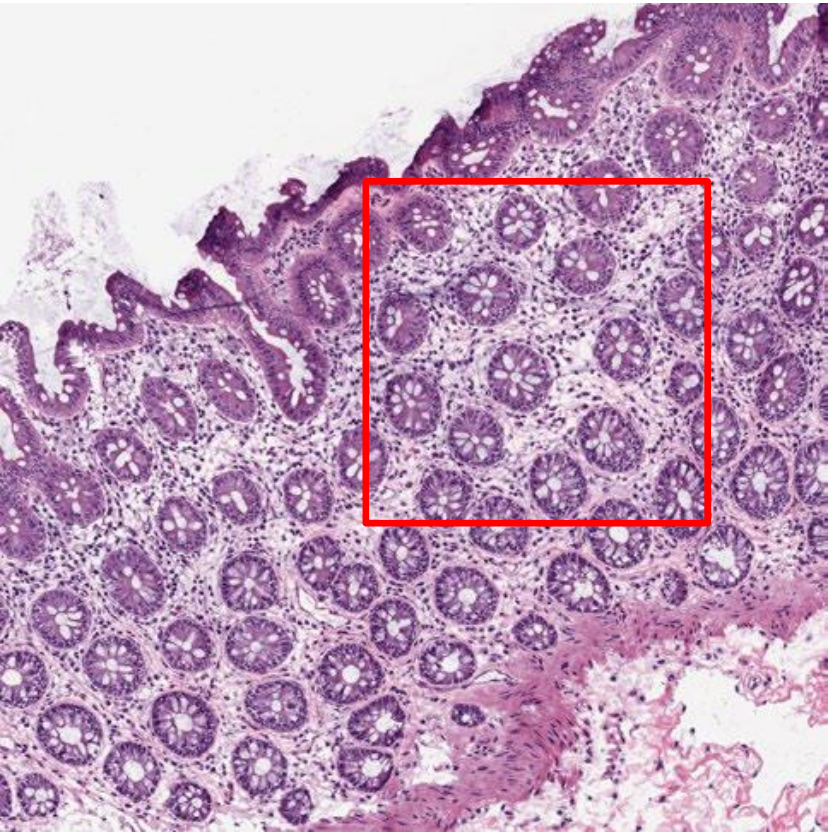
a **Original image (normal)**



non-neoplastic morphology

★ Prediction model

Original prediction

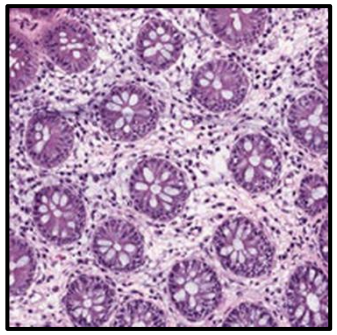
Normal

Tumor

b **Counterfactual question**

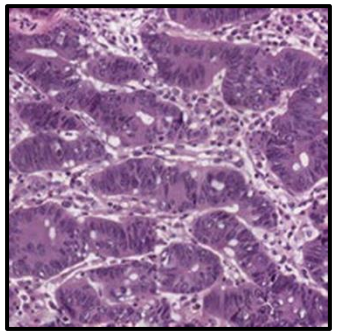
What counterfactual *change* to this tissue would shift the prediction from "Normal" to "Tumor"?

Original region from (a)



Organized glandular architecture

After intervention from (b)

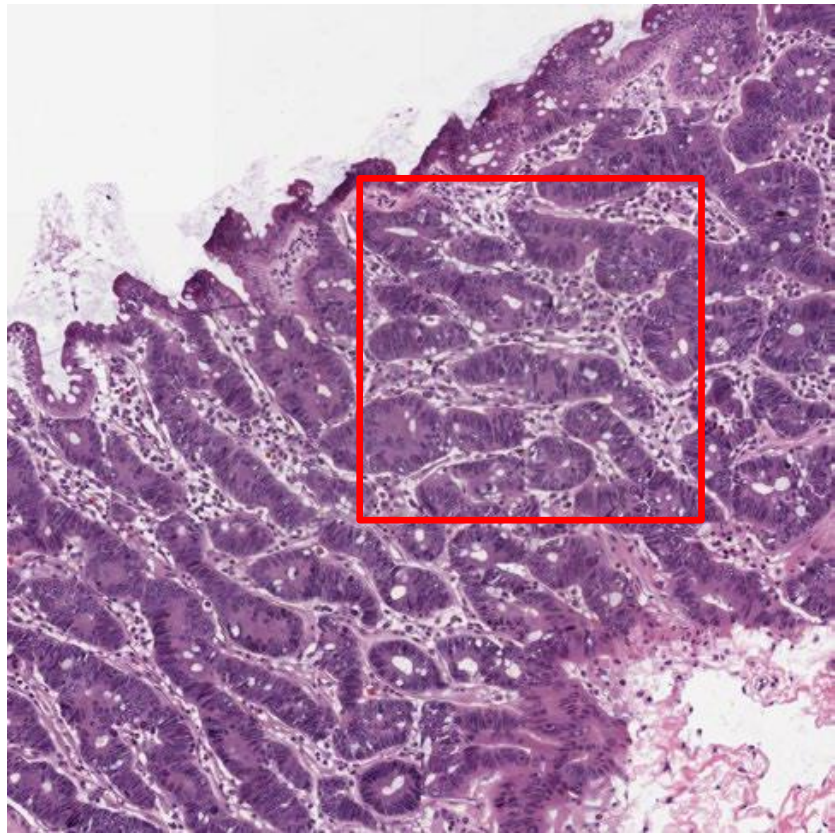


Altered glandular architecture

Change selected morphology while retaining the surrounding tissue context.

★ Same fixed prediction model

c **Generated counterfactual image (tumor)**



COAD | tumor-associated morphology

★ Prediction model

After counterfactual

Normal

Tumor